

Comparative Analysis of AegisLearn with Existing Models

This comparison examines the proposed AegisLearn framework against the five approaches included in the literature survey. The analysis focuses on the functional scope of each approach rather than reported numerical performance.

1. Comparative Analysis Table

Model / Study	Core Approach	RAG / Retrieval	Personalization	Adaptive Learning	Bloom's Taxonomy	Content Generation	Tutoring / Guidance	Main Limitation / Gap
GPT-4 + RAG in MOOCs (2024)	GPT-4 with retrieval-augmented generation for MOOC educational responses.	Yes	Limited	No	No	Yes	No	Improves response reliability, but learning-outcome evaluation and full personalization are limited.
LLMs as Virtual Tutors (2024)	Large language models used for tutoring, questioning and explanations.	No	Limited	Limited	No	Yes	Yes	Supports tutoring functions, but studies are relatively small-scale and domain coverage is limited.
GAIDE (2024)	Generative AI and prompt engineering for course-content creation.	No	No	No	No	Yes	Yes	Primarily supports content creation rather than a complete personalized learning cycle.
RAG for Knowledge-Intensive NLP (2020)	Retrieval-augmented generation using dense retrieval for knowledge-intensive tasks.	Yes	No	No	No	Yes	No	Establishes retrieval-grounded generation, but is not designed specifically for education or adaptive learning.
GenAI for Education (2024)	Review of generative AI / LLM applications in education.	No	Yes*	Yes*	No	Yes	Yes	Identifies personalization and adaptive-learning potential, but does not provide one unified implemented framework.
AegisLearn (Proposed)	RAG + prompt engineering + Bloom's Taxonomy + learner-profile-based adaptation.	Yes	Yes	Yes	Yes	Yes	Yes	Combines retrieval, personalization, adaptive question generation and learning support in one framework.

***Note:** For the GenAI-for-Education review, personalization and adaptive learning are treated as areas of potential/discussion in the literature, not as evidence that every reviewed system implements both features. The table is a qualitative scope comparison.

2. Graphical Comparative Analysis

The following graphs convert the qualitative comparison into a visual feature-coverage view. They are not accuracy, F1-score, or user-study performance graphs. They simply show which functional areas are addressed in the scope of each approach.

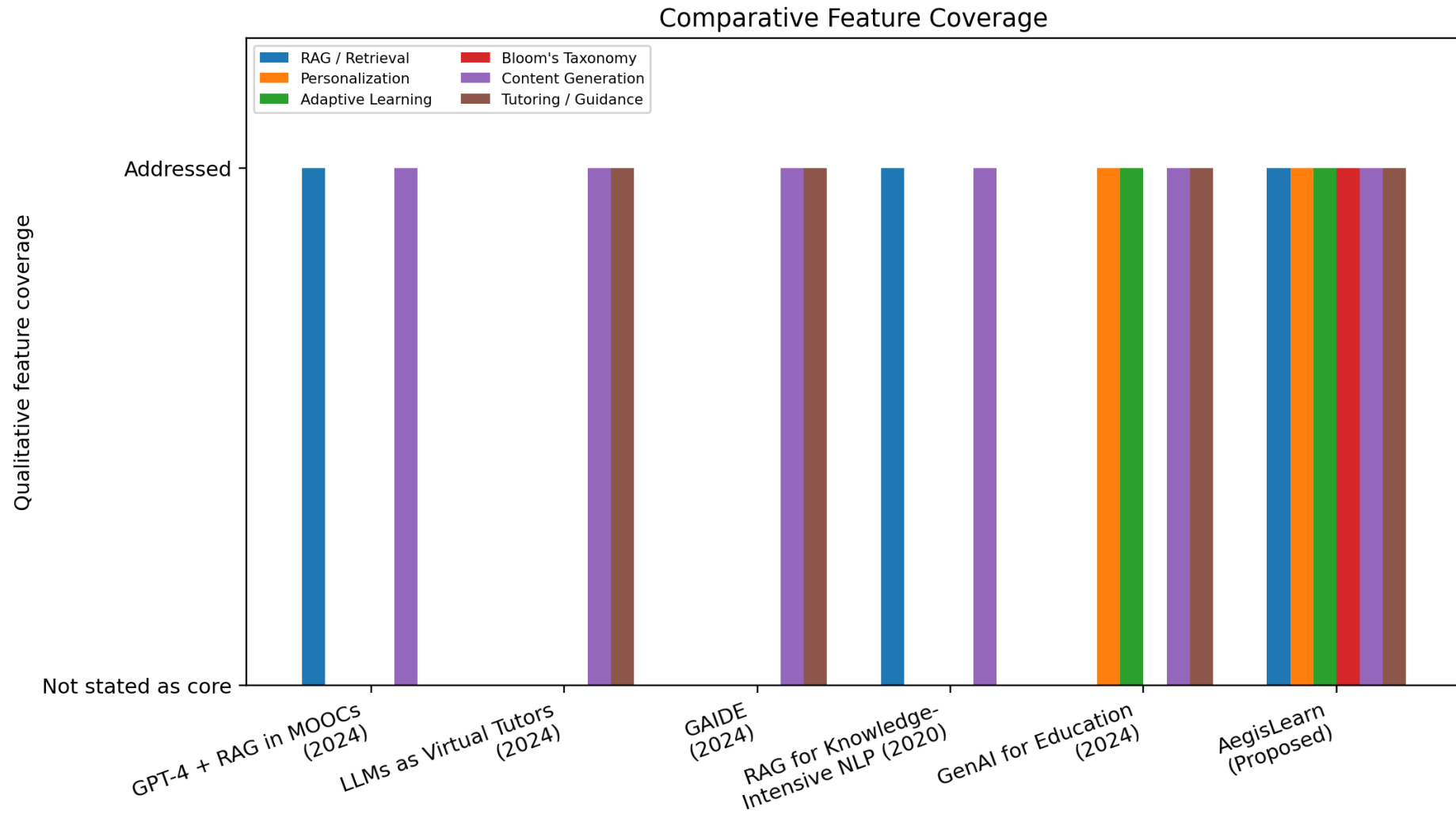


Figure 1. Qualitative feature coverage across the compared approaches.

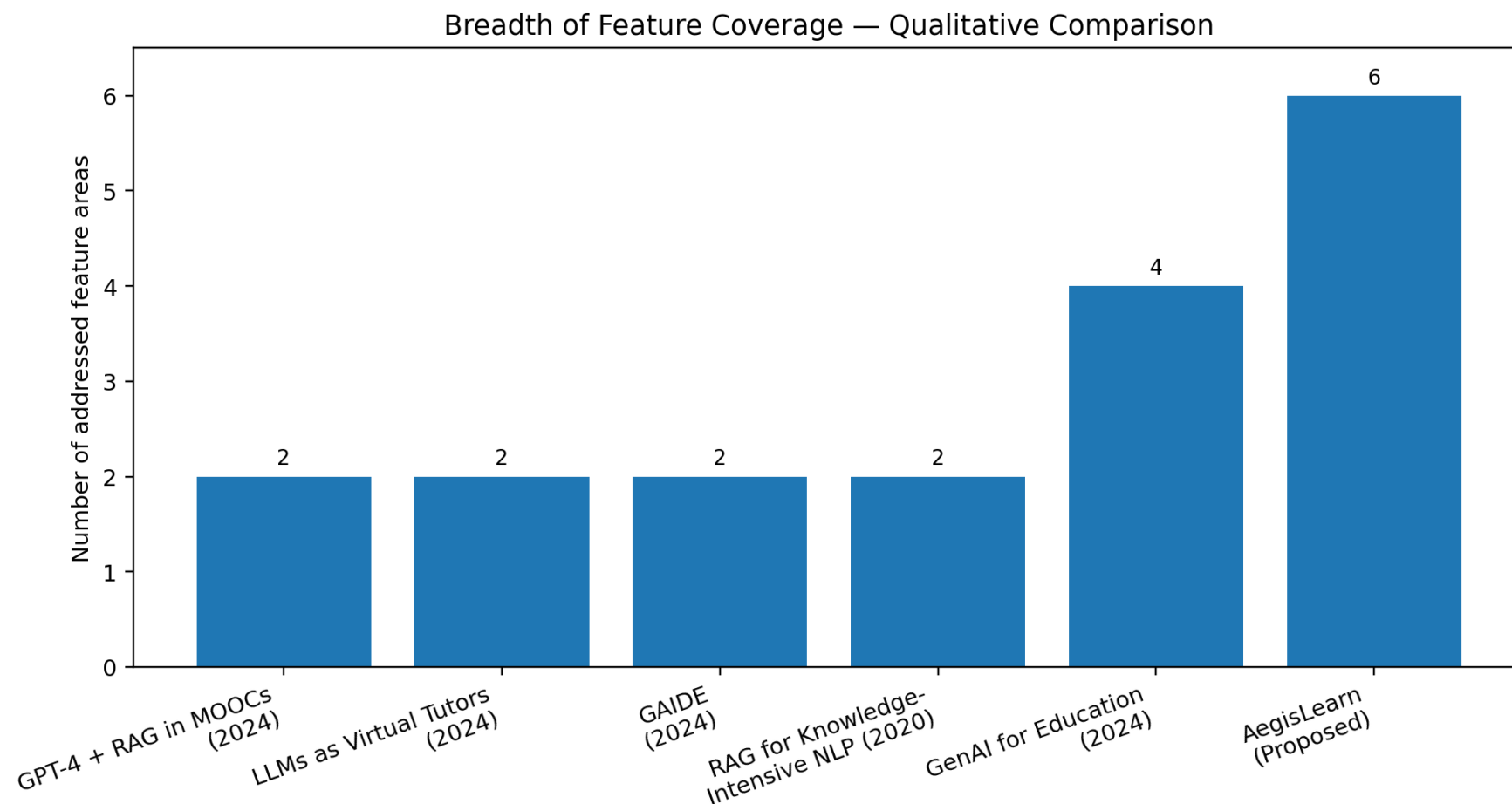


Figure 2. Breadth of addressed feature areas. AegisLearn covers the six feature areas defined for this comparative analysis; existing approaches address subsets of these areas.

3. Comparative Analysis / Research Gap

- Existing approaches demonstrate the usefulness of LLMs for educational content generation, tutoring, or retrieval-grounded responses.
- RAG-oriented approaches improve access to relevant external knowledge, but the general RAG work is not designed as a complete educational personalization framework.
- Virtual-tutor and course-content approaches provide interaction or generation, but do not necessarily combine retrieval, learner adaptation and structured cognitive-level assessment.
- The GenAI-for-Education literature identifies personalization and adaptive learning as important opportunities, while a unified implementation combining these components remains a gap.

- AegisLearn is proposed to address this integration gap by combining RAG, learner personalization, adaptive learning, Bloom's Taxonomy-based question generation, and educational content generation within one framework.

4. Conclusion of the Comparison

The comparison indicates that the main distinction of AegisLearn is the integration of multiple educational functions rather than reliance on a single model or technique. The proposed framework brings together retrieval-grounded generation, learner-specific adaptation, cognitive-level question generation and learning support. This provides the basis for the research gap and the proposed system design presented in the project.